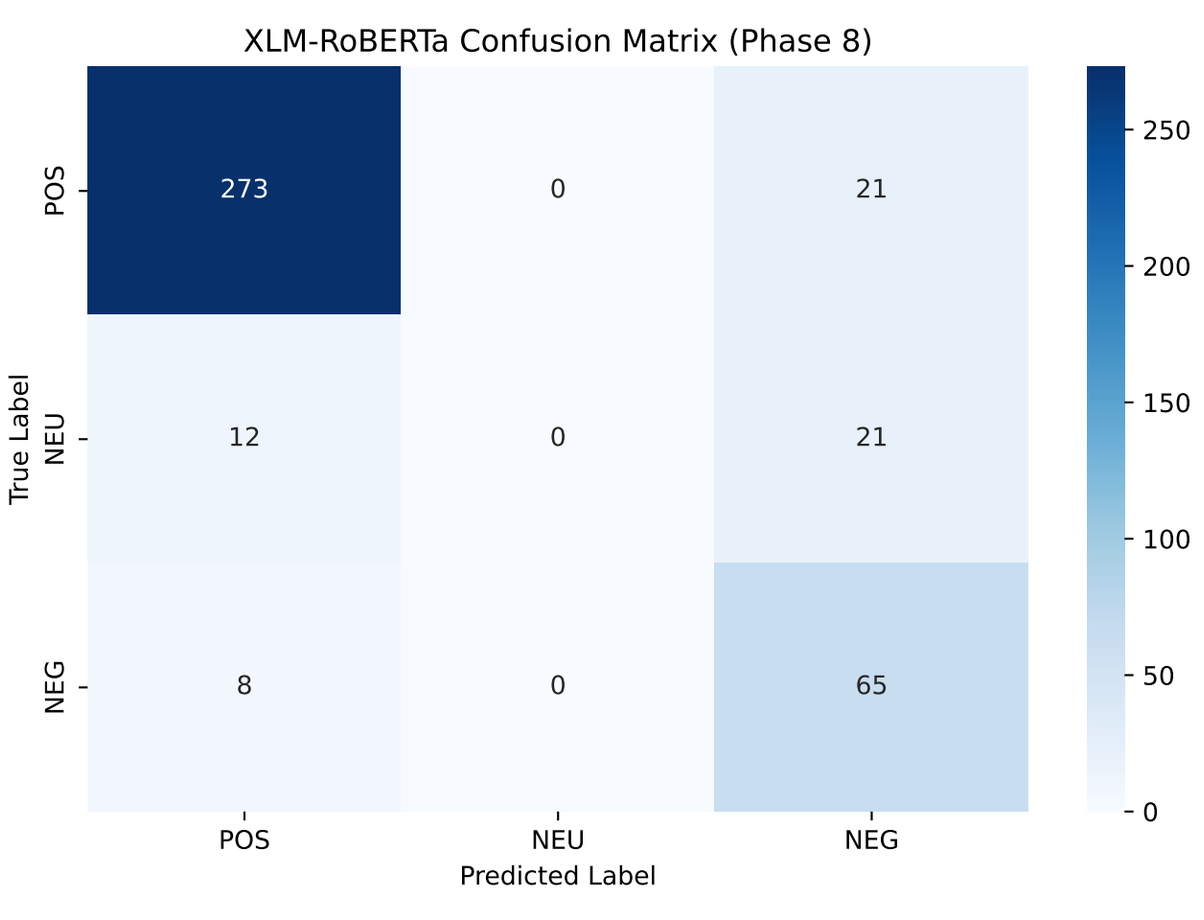
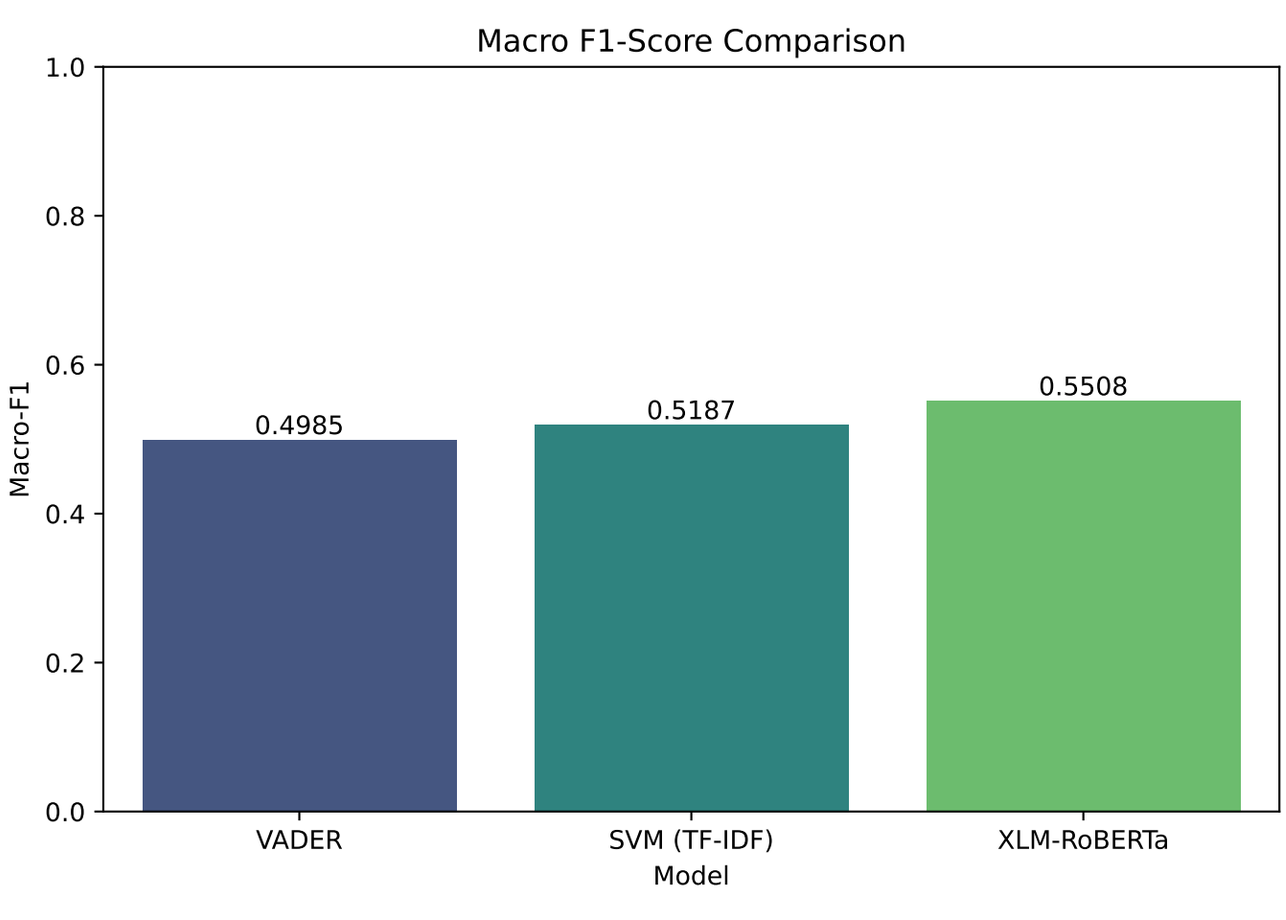

XLM-RoBERTa Confusion Matrix (Phase 8)
True Label
POS
NEU
NEG
273
0
21
12
0
21
8
0
65
250
200
150
100
50
0
POS
NEU
NEG
Predicted Label


Macro F1-Score Comparison
Macro-F1
1.0
0.8
0.6
0.4
0.2
0.0
0.4985
0.5187
0.5508
VADER
SVM (TF-IDF)
XLM-RoBERTa
Model

PHASE 8: OUT-OF-SAMPLE EVALUATION REPORT
1. FINAL BENCHMARK METRICS (Expanded)

Model	Accuracy	Precision (Macro)	Recall (Macro)	Macro-F1	F1 (POS)	F1 (NEU)	F1 (NEG)
VADER	0.6875	0.5685	0.5052	0.4985	0.8293	0.2034	0.4630
SVM (TF-IDF)	0.7300	0.5123	0.5290	0.5187	0.8502	0.1875	0.5185
XLM-RoBERTa	0.8450	0.5131	0.6063	0.5508	0.9302	0.0000	0.7222

2. MCNEMAR'S STATISTICAL TEST (alpha=0.05)

Baseline	XLM-R Correct (b)	Base Correct (c)	Chi-Square	p-value
VADER Lexicon	85	22	37.0935	1.1260e-09
SVM (TF-IDF)	58	12	30.2286	3.8401e-08

3. PHASE 7 GRID SEARCH SUMMARY

Batch Size	Learning Rate	Epochs	Status
16	2e-5	5	Suboptimal
16	3e-5	5	Suboptimal
16	5e-5	5	Suboptimal
32	2e-5	5	Suboptimal
32	3e-5	5	Optimal
32	5e-5	5	Suboptimal